# Supplementary Information (SI)

## Titania-Decorated Silicon Carbide Containing Cobalt Catalyst for the Fischer-Tropsch Synthesis

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**Section S1** Technical details of $^{59}Co$ zero field nuclear magnetic resonance ($^{59}Co$ NMR) on $10Co/TiO_2$-SiC catalyst

The NMR spectra taken at 2, 4.2, and 77 K of the sample $10Co/TiO_2$-SiC are given in Figure S5. As can been seen the NMR signal decreases with the increase of the measurement temperature. To explain this decrease we have to take in consideration that zero field NMR in ferromagnets is able to get a signal only if the magnetization of the sample is blocked (fixed in direction) during the timescale of the NMR measurement. For small Co particles as the one studied in this work the temperature below which the particle magnetization is blocked depends on the size of the particles: the smaller the particles the lower the blocking temperature. Therefore when the NMR measurement temperature is increased, the particles for which the blocking temperature is smaller than the measurement temperature will vanish from the NMR spectra. This explains why the NMR signals decreases when the measurement temperature is increased. Since the NMR measurement time scale is of the order of $10^{-0.5}$ s, it is possible to determine the minimum particle size measured at a given temperature by NMR. Assuming spherical particles, the minimum particle size is 4, 6, and 15 nm for measurement temperatures of 2, 4.2, and 77 K respectively. In addition, the NMR frequency also depends on the magnetic state of the particles. When the particles are small enough, the particle magnetic state is single domain this means that the magnetization points in the same direction all over the particle. In this case the NMR signal show contributions above 220 MHZ only. If the particle become larger than 60 nm a single domain magnetic state is not possible anymore. Therefore for such size of particle the magnetic state becomes multi-domain. This means that the magnetization in the particles is not homogeneous anymore but is divided into several magnetic domains with different magnetization direction. This results in a decrease of the NMR resonance frequency in the 216 MHz range. As can be seen in figure NMR 1 a significant NMR intensity is observed in the 216 MHz range showing that a large number of Co atoms are situated in particles with a diameter larger than 60 nm.

This sensitivity of the NMR measurement to the blocking temperature of the particles allows probing the Co atoms distribution versus the particles blocking temperature. Indeed, the NMR intensity difference between the spectra taken at different temperatures is directly proportional to the number of Co atoms situated in the respective blocking temperature ranges. This is summarized in fig. 8a. However this shows a distribution of Co atoms and not the Co particle size distribution that would be relevant for the discussion of the properties of the samples. To compute the particle size distribution we have to assume that the particles are spherical and we have to define an average particle size in each blocking temperature range: 5, 10.5 and 60 nm. The resulting particle size distribution is shown in Figure 8B. It might look surprising that about 70% of the Co atoms are included into less that 1% of large Cobalt particles. However we have to remember that the number of Co atoms included into a spherical particle increases as the cube of the particle size. Therefore a very small number of very large Co particles will involve easily a large number of Co atoms. That is why even thought only about 30% of the Co particles are involved into particles smaller than 15 nm, more that 99% of the particles found in these samples have a size smaller than 15nm.

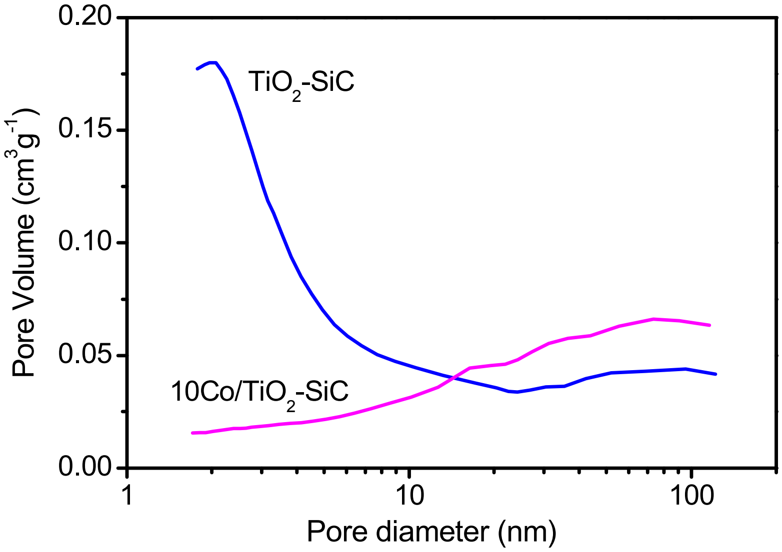


**Figure S1**. BJH pore size distribution of $TiO_2$-SiC support and 10Co/$TiO_2$-SiC catalyst.

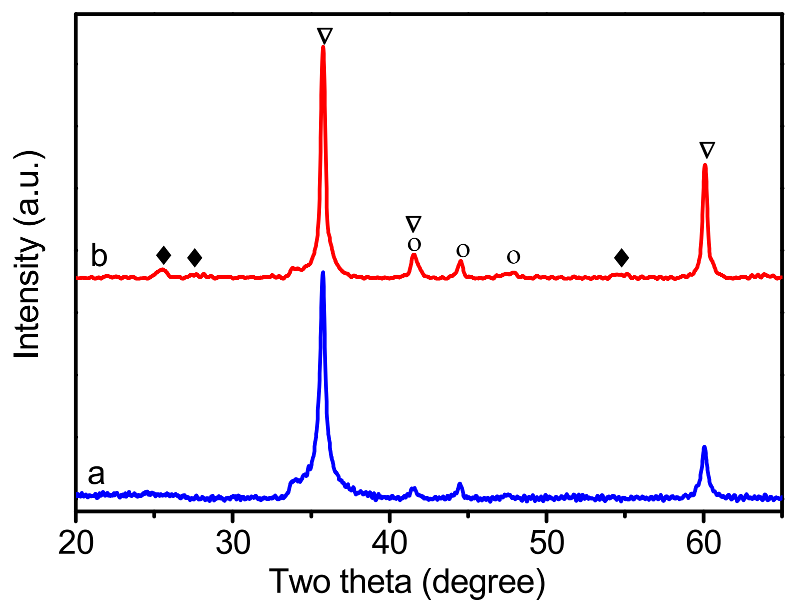


**Figure S2**. Ex-situ X-ray diffraction patterns of the catalysts: (a) 10Co/SiC; (b) 10Co/$TiO_2$-SiC. The catalysts were calcined at 350 $^{\circ}$C for 2 h followed by reduction under $H_2$ at 300 $^{\circ}$C for 6 h. Legends: Diamond (♦): $TiO_2$; circles (○): Co; triangle (▽): SiC.

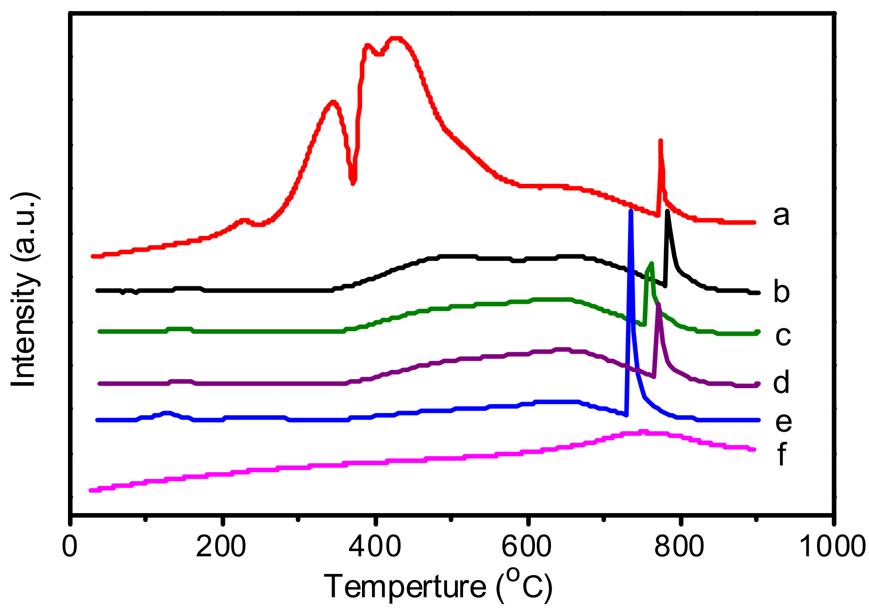


**Figure S3.** (a) The TPR of 10Co/$TiO_2$-SiC was directly test under diluted $H_2$ with heating rate of 15 °C $min^{-1}$; sequential TPR of 10Co/$TiO_2$-SiC sample was reduced by diluted $H_2$ (10 vol. % in argon**)** at 300 °C for (b) 2h; (c) 3h; (d) 4h; (e) 6h in TPR equipment, and then reduced under the diluted $H_2$ with heating rate of 15 °C $min^{-1}$. (f) TPR profile of $TiO_2$-SiC support.

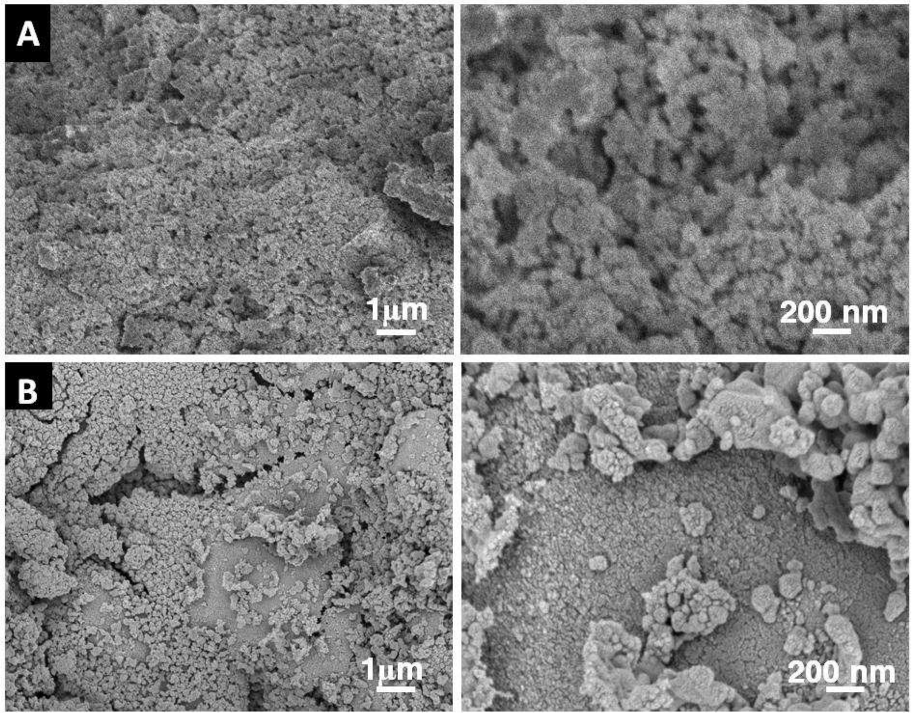


**Figure S4.** (A) SEM micrographs of the 10Co/SiC after reduction under $H_2$ at 300 °C for 6 h. (B) SEM micrographs of the 10Co/$TiO_2$-SiC after reduction under $H_2$ at 300 °C for 6 h.

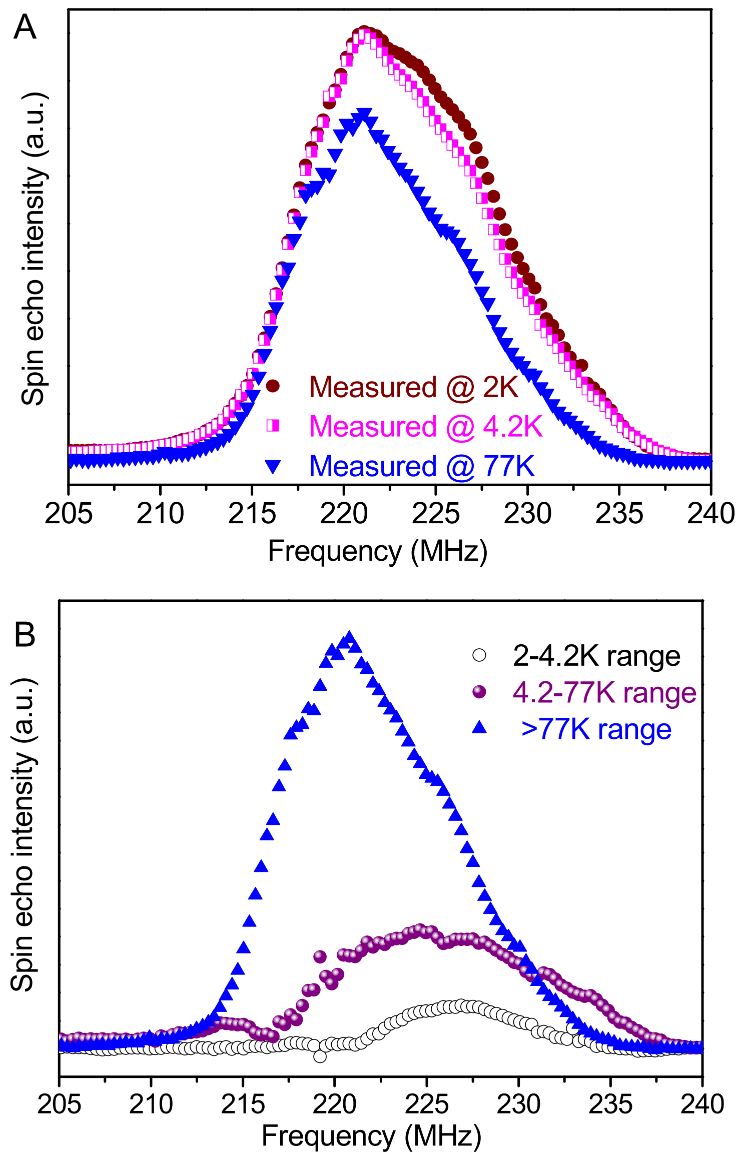


**Figure S5** $^{59}$Co NMR spectra of the 10Co/$TiO_2$-SiC catalyst (A) recorded at 2K, 4.2K and 77K, respectively; (B) analyzed in different blocking temperature range.

**Table S1.** Fischer-Tropsch results obtained on 10 wt% and 30 wt% cobalt on Titanium oxide decorated SiC-based. Reaction conditions: GHSV = 2850 $h^{-1}$, $H_2/CO$ = 2, pure syngas, total pressure = 40 bar.

| Catalyst | T(°C) | CO Conversion (%) | Product selectivity (%) | | | | FTS rate# |
|---|---|---|---|---|---|---|---|
| | | | $CO_2$ | $CH_4$ | $C_2$–$C_4$ | $C_{5+}$ | |
| $10Co/TiO_2$-SiC | 215 | 33.9 | 0 | 3.2 | 1.5 | 95.3 | 0.24 |
| | 220 | 37.0 | 0 | 3.9 | 1.7 | 94.4 | 0.26 |
| $30Co/TiO_2$-SiC | 215 | 57.7 | 3.1 | 4.7 | 0 | 92.2 | 0.40 |
| | 220 | 68.3 | 2.7 | 4.7 | 0 | 92.5 | 0.47 |

# FTS rate, $g_{C5+} \cdot g_{catalyst}^{-1} \cdot h^{-1}$ (Mass $C_{5+}$ produced per g catalyst per hour)

**Table S2.** Fischer-Tropsch synthesis performance under more severe reaction conditions. Reaction conditions: reaction temperature = 230 °C, GHSV = 3800 $h^{-1}$, $H_2/CO$ = 2, pure syngas, total pressure = 40 bar.

| Catalyst | CO Conversion (%) | Product selectivity (%) | | | | FTS rate# |
|---|---|---|---|---|---|---|
| | | $CO_2$ | $CH_4$ | $C_2$–$C_4$ | $C_{5+}$ | |
| $30Co/TiO_2$-SiC | 61.8 | 3.1 | 6.2 | 0.1 | 90.6 | 0.56 |
| 30Co-Ru/SiC-E [Ref.] | 47.0 | 3.0 | 5.5 | 0.0 | 91.5 | 0.43 |

# FTS rate, $g_{C5+} \cdot g_{catalyst}^{-1} \cdot h^{-1}$ (Mass $C_{5+}$ produced per g catalyst per hour)

Reference: de Tymowski, B.; Liu, Y. F.; Meny, C.; Lefevre, C.; Begin, D.; Nguyen, P.; Pham, C.; Edouard, D.; Luck, F.; Pham-Huu, C. *Appl Catal a-Gen* **2012**, *419*, 31.